

Multiple Choice (10 marks)

1. A digital signature needs a
 - (a) Private-key system
 - (b) Shared-key system
 - (c) Public-key system
 - (d) All of them
2. Exploitation of the Heartbleed bug permits
 - (a) overwriting cryptographic keys in memory
 - (b) a kind of code injection
 - (c) a read outside bounds of a buffer
 - (d) a format string attack
3. Failed sessions allow brute-force attacks on access credentials. This type of attacks are done in which layer of the OSI model?
 - (a) Physical layer
 - (b) Data-link Layer
 - (c) Session layer
 - (d) Presentation layer
4. SHA-1 has a message digest of
 - (a) 160 bits
 - (b) 512 bits
 - (c) 628 bits
 - (d) 820 bits
5. How is IP address spoofing detected?
 - (a) Installing and configuring a IDS that can read the IP header
 - (b) Comparing the TTL values of the actual and spoofed addresses
 - (c) Implementing a firewall to the network
 - (d) Identify all TCP sessions that are initiated but does not complete successfully

True / False (10 marks)

Answer True or False

6. True or False: Authentication is a process used to verify the identity of a user and is not classified as a security exploit.
7. True or False: A MAC with a fixed output length can still be secure if the underlying cryptographic functions are strong.
8. True or False: The length of the message digest in MD 5 is 160 bits.
9. True or False: A session symmetric key between two parties is used only once for secure communication during that session.
10. True or False: A crypter is a type of malware that disguises itself as a legitimate program to avoid detection by antivirus software.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find element at a given index after number of rotations.
12. Write a python function to count the number of lists in a given number of lists.

End of Paper